Anchored Cascade Control with Conditional Activation for Wheeled Bipedal Balance and Disturbance Recovery

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***Abstract*—Wheeled bipedal robots must simultaneously achieve millimeter-level standing precision, omnidirectional disturbance recovery, and operation across uneven terrain—capabilities that create fundamental trade-offs in controller design. Stiffness that holds sub-millimeter idle precision shrinks the capture basin for push recovery; integral action that cancels equilibrium bias winds up during large disturbances and prevents recovery. We present the Anchored Cascade Controller (ACC), a unified classical control architecture in which five components are activated continuously by smoothstep gates on physical conditions (proximity, quietness, contact, terrain) rather than by discrete switching. The core contribution is a proximity-gated anchor mechanism: a position integral confined to a $15\,\mathrm{cm}$ neighborhood around the home position, gated by an asymmetric envelope follower (attack $\tau \approx 30\,\mathrm{ms}$, release $\tau \approx 1.5\,\mathrm{s}$) that distinguishes true quiet stance from post-push ringdown. Coupled with scheduled pitch stiffness and a gated damping boost, this mechanism resolves a fundamental trade-off: baseline controllers without the anchor achieve comparable push survival ($F_{\min} = 80 - -84\,\mathrm{N}$ across all configurations) but exhibit $55\,\mathrm{mm}$ RMS idle limit cycles—a $180\times$ degradation in standing precision. ACC is the only configuration that delivers both mm-level idle ($0.3\,\mathrm{mm}$ RMS) and competitive omnidirectional push recovery ($F_{\min} = 80\,\mathrm{N}$, $F_{\text{med}} = 113\,\mathrm{N}$, 8-direction polar sweep at $400\,\mathrm{Nm/s}$ torque rate limit). The anchor mechanism does not increase peak push survival—all configurations cluster at $F_{\min} \approx 80\,\mathrm{N}$—rather, it eliminates the equilibrium bias that sustains perpetual oscillation in P-only controllers, and the gated damping boost ensures post-push ringdown converges to zero within $9\,\mathrm{s}$ instead of persisting indefinitely. The architecture also includes load-gated flight attitude control for contact-loss recovery (drop $100\,\mathrm{cm}$, ledge drive-off $50\,\mathrm{cm}$) and per-leg proprioceptive terrain adaptation (one-wheel curb $20\,\mathrm{cm}$). All behaviors are achieved without learning, training, or exteroceptive sensing.**

***Index Terms*—Wheeled biped robot, cascade control, anchored standing, proximity-gated integral, push recovery, flight recovery, terrain adaptation.**

I. Introduction

Wheeled biped robots combine the energy efficiency of wheel-based locomotion with the postural adaptability of articulated legs. However, controlling such platforms requires simultaneously satisfying three demanding objectives: millimeter-level standing precision at the commanded pose, omnidirectional recovery from substantial push disturbances, and operation across uneven terrain where the two wheels rest at different heights. Existing approaches address subsets of these challenges—whole-body control (WBC) for posture regulation [2], linear quadratic regulation (LQR) for balance near an operating point [1], model predictive control (MPC) for terrain negotiation [5]—but no single classical controller has demonstrated all three capabilities in a unified design.

The difficulty is not merely one of integration but of fundamental tension. On the stiffness axis, a single fixed pitch gain cannot occupy both ends: $k_p = 50$ holds $0.3\,\mathrm{mm}$ idle precision but shrinks the omnidirectional capture basin to $F_{\min} = 30\,\mathrm{N}$, while $k_p = 35$ widens capture but lets idle oscillate ($55\,\mathrm{mm}$ RMS). On the integral axis, the same dilemma recurs: omitting integral action leaves a $1.3\,\mathrm{Nm}$ equilibrium bias that sustains a perpetual $\pm 4 - -6\,\mathrm{cm}$ limit cycle, yet a global integral winds up during a push and blocks recovery. These are intrinsic trade-offs—any single-parameter choice sacrifices one axis.

We present the Anchored Cascade Controller (ACC), which resolves both trade-offs through a single design principle: *conditional activation*. Rather than selecting one gain or enabling one mechanism globally, ACC activates each control component continuously via smoothstep gates on physical conditions. The core mechanism is a **proximity-gated anchor**: a position integral confined to the $15\,\mathrm{cm}$ neighborhood around home, gated by an **asymmetric envelope follower** (fast-attack $\tau \approx 30\,\mathrm{ms}$, slow-release $\tau \approx 1.5\,\mathrm{s}$) that detects true quiet stance without being fooled by post-push ringdown. Coupled with **scheduled pitch stiffness** and a **gated damping boost**, ACC achieves $F_{\min} = 80\,\mathrm{N}$, $F_{\text{med}} = 113\,\mathrm{N}$ while maintaining $0.3\,\mathrm{mm}$ idle RMS. Critically, all tested configurations—including baselines without the anchor—cluster at $F_{\min} \approx 80\,\mathrm{N}$; the anchor mechanism does not raise the peak push threshold but rather eliminates the $55\,\mathrm{mm}$ idle limit cycle that afflicts every other configuration. ACC is the only configuration that delivers both mm-level standing and competitive disturbance rejection, with post-push ringdown converging to zero within $9\,\mathrm{s}$. The architecture further includes load-gated flight attitude control for contact-loss scenarios and per-leg proprioceptive ground adaptation for uneven terrain—all without learning, training, or exteroceptive sensing.

The main contributions of this work are:

1) A **proximity-gated anchor mechanism** with asymmetric envelope follower that confines integral action to the anchor neighborhood, resolving the integral windup-vs-bias trade-off.
2) A **unified gate-structured two-channel cascade architecture** in which five control components are activated

continuously by smoothstep gates on physical conditions, with explicit wheel-torque and leg-joint-torque channel separation.

3) **Flight recovery and per-leg terrain adaptation** without exteroception: load-gated reaction-wheel attitude control for contact loss and contact-point-based ground estimation for uneven terrain.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the robot platform and motivates gain scheduling via TWIP pole migration. Section IV presents the ACC formulation. Section V provides experimental validation with ablation analysis. Section VI discusses limitations, and Section VII concludes.

II. Related Work

A. Wheeled Bipedal Platforms

Wheeled bipedal robots have been explored through several platforms over the past decade. Ascento [1] demonstrated a two-wheeled jumping robot with LQR-based balancing on flat ground; subsequent work extended this to whole-body control with kinematic loops [2]. DIABLO [3] introduced a six-DOF direct-drive design with cascaded LQR for height-variable stance. Whleaper [4] is a ten-DOF platform most similar to ours in morphology, using both LQR and PPO for multimodal locomotion. SKATER [5] demonstrated MPC-based turning robustness and terrain adaptability on a wheeled biped. These platforms rely on fixed-gain LQR [1], [3], [4], whole-body MPC [5], learned policies [4], or adaptive WBC [8]—none provide a single classical controller integrating anchored standing, omnidirectional push recovery, contact-loss control, and per-leg terrain adaptation. Feng *et al.* [9] combined LQR with active disturbance rejection control (ADRC) for a wheel-legged robot on bumpy terrain, the closest prior attempt at disturbance-aware classical control for this morphology; however, their approach addresses terrain disturbances through estimation rather than conditional mechanism activation and does not handle contact loss or uneven terrain.

B. Cascade Control and Conditional Activation

Cascade control structures are standard in industrial process control [10] and have been applied to legged robots for hierarchical torque regulation. However, these typically use fixed-gain inner-outer loops without conditional activation gates. Closest to our approach, gain-scheduled control [11] varies controller parameters with a measured scheduling variable (e.g., airspeed in flight control), but the schedule is typically a continuous function of a single variable, not a multi-gate conditional structure responding to qualitatively distinct physical regimes (near-home quiet stance vs. displaced ballistic recovery vs. contact loss).

C. Contact-Loss and Flight-Phase Control

Handling contact loss is critical for wheeled bipeds traversing ledges or dropped from height. Prior work on bipedal flight-phase control [12] uses pre-planned trajectories and impact mitigation based on capture-point dynamics. Ruscelli *et al.* [7] demonstrated dual-flywheel orientation control for quadrupedal flight phases, using conservation of angular momentum for landing reorientation from 30° pitch drops—the closest prior work to our reaction-wheel flight strategy. Chamorro *et al.* [6] recently demonstrated blind stair climbing on the Ascento platform using asymmetric actor-critic RL, achieving $15\,\mathrm{cm}$ step negotiation with sim-to-real transfer. However, continuous load-gated flight engagement with asymmetric hysteresis has not been demonstrated for wheeled bipedal ledge descent. Our flight PD uses the drive wheels as reaction flywheels during contact loss, with a 5-step release hysteresis and $150\,\mathrm{ms}$ soft handback to prevent landing-bounce re-launch.

D. Proprioceptive Terrain Adaptation

Terrain adaptation in legged robots typically relies on exteroception (depth cameras, lidar) [13] or whole-body MPC replanning [5]. Blind proprioceptive approaches estimate ground height from joint torques or contact forces [14]. ACC extends this to per-leg independent adaptation: contact-point force measurements estimate ground height separately for each wheel, and the height command is split per leg with closed-loop leveling—requiring no terrain model and no exteroception.

III. Robot Platform and TWIP Model

A. Robot Morphology

The robot, shown in Fig. 1, is a ten-DOF wheeled biped with five actuated joints per leg: hip roll (HR), hip yaw (HY), hip pitch (HP), knee (KN), and drive wheel (WH). The platform has a total mass of $8.1\,\mathrm{kg}$, thigh length $0.26\,\mathrm{m}$, shin length $0.28\,\mathrm{m}$, wheel radius $0.06\,\mathrm{m}$, and hip width $0.23\,\mathrm{m}$. Actuator torque limits range from $60\,\mathrm{Nm}$ (HR, HY, WH) to $150\,\mathrm{Nm}$ (HP, KN). Sensors include one IMU (accelerometer, gyroscope) and ten joint encoders.

B. Simulation Environment

All experiments use the MuJoCo physics engine [15] at $500\,\mathrm{Hz}$ ($\Delta t_{\text{sim}} = 0.002\,\mathrm{s}$) with the controller running at $100\,\mathrm{Hz}$ ($\Delta t_{\text{ctrl}} = 0.01\,\mathrm{s}$, ten simulation substeps per control step). The controller outputs raw torque commands applied directly to MuJoCo actuators; no intermediate position or velocity servo is used.

C. TWIP Model and Motivation for Gain Scheduling

The sagittal balance problem is modeled as a two-wheeled inverted pendulum (TWIP). Linearizing about the upright equilibrium with CoM height l_c yields the unstable open-loop pole

$$\lambda_u = \sqrt{\frac{g}{l_c}}, \tag{1}$$

where $g = 9.81\,\mathrm{m/s^2}$. As the commanded CoM height varies across the operating range $l_c \in [0.40\,,0.70\,]\,\mathrm{m}$, this pole migrates from $3.74\,\mathrm{rad/s}$ to $4.95\,\mathrm{rad/s}$—a 32% variation. No single fixed-gain LQR covers this range: gains tuned at

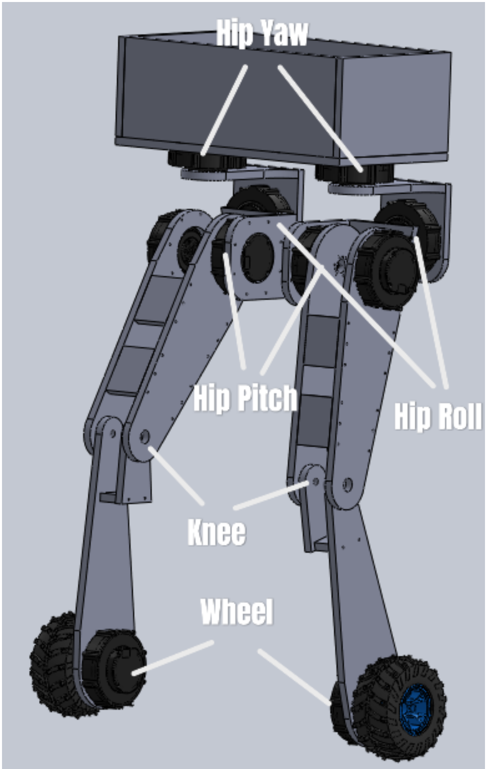

Fig. 1. Robot joint layout. Each leg: hip roll (HR), hip yaw (HY), hip pitch (HP), knee (KN), wheel (WH).

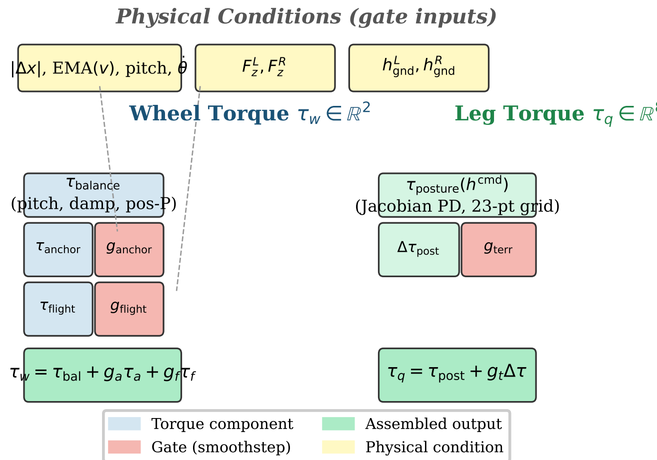

Fig. 2. ACC two-channel torque assembly. Physical conditions (top) gate each component via continuous smoothstep functions (dashed). The two channels—wheel torque $\boldsymbol{\tau}_w$ and leg-joint torque $\boldsymbol{\tau}_q$—terminate independently at the actuators.

the nominal height ($l_c = 0.54\,\text{m}$, $\lambda_u \approx 4.26\,\text{rad/s}$) are insufficiently stiff at low postures and over-damped at high postures.

Motivational scope. The pole-migration argument is purely motivational: it explains why a single fixed gain fails, not how ACC's gains are obtained. ACC does not solve an LQR; its scheduled gains are proportional-derivative terms tuned per height band. A standard LQR appears only as a comparison baseline in Section V.

IV. Unified ACC Formulation

A. Two-Channel Torque Assembly

ACC is a single controller whose five components are assembled into two independent torque channels—wheel torque ($\boldsymbol{\tau}_w \in \mathbb{R}^2$) and leg-joint torque ($\boldsymbol{\tau}_q \in \mathbb{R}^8$)—as shown in Fig. 2:

$$\begin{aligned} \boldsymbol{\tau}_w &= \boldsymbol{\tau}_{\text{balance}} + g_{\text{anchor}} \cdot \boldsymbol{\tau}_{\text{anchor}} + g_{\text{flight}} \cdot \boldsymbol{\tau}_{\text{flight}}, \\ \boldsymbol{\tau}_q &= \boldsymbol{\tau}_{\text{posture}}(h^{\text{cmd}}) + g_{\text{terrain}} \cdot \Delta\boldsymbol{\tau}_{\text{posture}}(h^{\text{cmd},L},\, h^{\text{cmd},R}). \end{aligned} \tag{2}$$

Each gate g_* is a **continuous smoothstep function** of physical conditions (proximity, quietness, contact forces, terrain disparity)—not a discrete switch. The two channels are assembled independently: wheel torques control sagittal balance, lateral stabilization, and yaw; leg-joint torques control posture regulation and per-leg height adaptation. This separation is principled: the sagittal balance problem is fundamentally a wheel-control problem, while posture is a leg-joint-control problem.

B. Sagittal Balance Core ($\boldsymbol{\tau}_{balance}$)

The sagittal balance component is always active and acts entirely in the wheel-torque channel:

$$\boldsymbol{\tau}_{\text{balance}} = \begin{bmatrix} \tau_{\text{pitch}} + \tau_{\text{damping}} + \tau_{\text{pos}} \\ \tau_{\text{pitch}} + \tau_{\text{damping}} + \tau_{\text{pos}} \end{bmatrix} + \begin{bmatrix} +\tau_{\text{lat}} + \tau_{\text{yaw}} \\ -\tau_{\text{lat}} - \tau_{\text{yaw}} \end{bmatrix}, \tag{3}$$

where common-mode terms ($\tau_{\text{pitch}}, \tau_{\text{damping}}, \tau_{\text{pos}}$) are applied symmetrically and lateral/yaw terms differentially. The individual terms are:

$$\tau_{\text{pitch}} = k_p(h, v) \cdot \theta + k_d \cdot \dot{\theta}_{\text{notch}}, \tag{4}$$

$$\tau_{\text{damping}} = -k_{\text{vel}} \cdot v_{\text{sag}}, \tag{5}$$

$$\tau_{\text{pos}} = -k_{\text{pos}} \cdot \text{clip}(\Delta x,\, \pm 0.10\,\text{m}), \tag{6}$$

$$\tau_{\text{lat}} = k_{p,\text{roll}} \cdot \phi + k_{d,\text{roll}} \cdot \dot{\phi}, \tag{7}$$

$$\tau_{\text{yaw}} = k_{p,\text{yaw}} \cdot e_\psi + k_{d,\text{yaw}} \cdot \omega_z, \tag{8}$$

where θ is the torso pitch angle, $\dot{\theta}_{\text{notch}}$ is the pitch rate after a biquad notch filter centered at 2.5 Hz, v_{sag} is the sagittal body velocity, Δx is the sagittal position error from the latched home position, ϕ is the roll angle, and e_ψ is the yaw error.

Notch filter. The 2.5 Hz notch removes a closed-loop standing oscillation mode induced by the proportional position term acting on the wheeled-inverted-pendulum dynamics. This frequency was identified from the FFT of the pitch signal under P-only standing, which exhibits a dominant peak at 2.5 Hz (Fig. 3 inset); it is distinct from the open-loop WIP pole at $\sqrt{g/l_c} \approx 0.7\,\text{Hz}$ (Section III-C). The biquad notch filter has quality factor $Q = 2.0$ and is height-gated (full notch only above $h = 0.47\,\text{m}$, blended below) to avoid over-attenuating the slower dynamics at low postures.

P-only position and its limitation. The proportional position term (k_{pos}, clipped to $\pm 0.10\,\text{m}$ error, saturating at $\pm 4\,\text{Nm}$) is sufficient to keep the robot within $\pm 4 - -6\,\text{cm}$ of the home position. However, P-only control creates an equilibrium in which $\tau_{\text{pos}} \approx 1.3\,\text{Nm}$ cancels the untrimmed pitch bias—the robot parks offset from home and a $\pm 4 - -6\,\text{cm}$ limit cycle persists indefinitely. This motivates the anchor integral, which cancels the bias without the windup risk of a global integrator.

Scheduled pitch stiffness. The pitch gain k_p transitions continuously between a stiff anchor value ($k_p = 50$, for sub-millimeter idle precision) and a soft recovery value ($k_p = 35$, which widens the capture basin):

$$k_p(\text{EMA}, \Delta x) = 50 - 15 \cdot g_{\text{kp}}(\text{EMA}, \Delta x), \tag{9}$$

where $g_{\text{kp}} = 1 - g_{\text{env}} \cdot g_{\text{prox}}$ is a composite gate that approaches 0 at anchor (quiet, near home $\rightarrow k_p = 50$) and 1 during ballistic recovery (oscillating or displaced $\rightarrow k_p = 35$). This uses the same g_{env} (Eq. 14) and g_{prox} (Eq. 13) gates as the anchor mechanism—the pitch stiffness softens smoothly as the robot leaves the anchor neighborhood or begins oscillating, and stiffens smoothly as it returns. Because g_{env} is driven by the slow-release envelope, k_p cannot modulate at the $2 - -3\,\text{Hz}$ cycle frequency, avoiding parametric pumping.

C. Proximity-Gated Anchor Mechanism ($g_{anchor} \cdot \boldsymbol{\tau}_{anchor}$)

The anchor mechanism is the core contribution. It adds two terms to the wheel-torque channel that are active only when the robot is near home *and* genuinely at rest:

$$\boldsymbol{\tau}_{\text{anchor}} = K_i \int g_{\text{prox}} \cdot g_{\text{env}} \cdot (-\Delta x)\, dt \; + \; g_{\text{boost}} \cdot K_{\text{boost}} \cdot (-v_{\text{sag}}), \tag{10}$$

where all gates are continuous smoothstep functions designed to equal 1 when the robot is "at home and quiet" and transition smoothly to 0 otherwise:

$$\begin{aligned} g_{\text{prox}}(\Delta x) &= 1 - \text{smoothstep}(|\Delta x|,\ 0.05,\ 0.15), \\ g_{\text{env}}(v_{\text{sag}}) &= 1 - \text{smoothstep}\big(\text{EMA}(|v_{\text{sag}}|), \\ &\qquad 0.18,\ 0.30\big), \end{aligned} \tag{11}$$

$$\begin{aligned} g_\theta(\theta, \dot\theta) &= \text{smoothstep}(|\theta|,\ 2^\circ,\ 12^\circ) \\ &\quad \cdot \text{smoothstep}(|\dot\theta|,\ 2^\circ/s,\ 15^\circ/s), \end{aligned} \tag{12}$$

$$g_{\text{boost}} = g_{\text{prox}} \cdot g_{\text{env}} \cdot g_\theta. \tag{13}$$

The pitch stability gate g_θ closes when either pitch or pitch rate exceeds its band, preventing the damping boost from fighting a gentle sustained push that does not trigger the velocity envelope. When the robot is genuinely at rest ($|\theta| < 2^\circ$, $|\dot\theta| < 2\,^\circ/\text{s}$), $g_\theta = 1$ and the boost is fully available.

Asymmetric envelope follower. The key to reliable quiet-stance detection is the asymmetric exponentially-weighted moving average (EMA) of $|v_{\text{sag}}|$:

$$\text{EMA}_t = \begin{cases} \alpha_a |v_t| + (1 - \alpha_a)\text{EMA}_{t-1}, & \text{if } |v_t| > \text{EMA}_{t-1}, \\ \alpha_r |v_t| + (1 - \alpha_r)\text{EMA}_{t-1}, & \text{otherwise}, \end{cases} \tag{14}$$

with attack coefficient $\alpha_a = 0.35$ ($\tau \approx 30\,\text{ms}$) and release coefficient $\alpha_r = 0.007$ ($\tau \approx 1.5\,\text{s}$). This asymmetry is essential: a symmetric EMA with comparable smoothing is *bistable*—a post-push oscillation sustained by the balanced limit cycle holds the EMA in a middle band where the gate is half-open and the boost is too weak to collapse the cycle. The fast-attack/slow-release envelope jumps to 1 (quiet) only when oscillation truly decays, and drops to 0 within $\sim 30\,\text{ms}$ of a disturbance. This behavior is validated by the symmetric-EMA ablation in Section V.

Why separate P (Section IV-B) and I (this section)? The P term is global and always active: it pulls the robot toward home from any distance. The I term is local and gated: it accumulates only within the anchor neighborhood to cancel the equilibrium bias, and is frozen (by $g_{\text{prox}} = 0$) outside $15\,\text{cm}$ to prevent windup. Both terms act in the *wheel-torque channel*—the bias originates from the wheel-level force balance, and correcting it at the wheels eliminates the limit cycle at its source rather than compensating through posture shift.

D. Posture and Yaw Stability ($\boldsymbol{\tau}_{posture}$)

Posture regulation acts in the leg-joint channel and uses Jacobian-based PD gains scheduled across a 23-point height grid to maintain consistent foot-level stiffness:

$$\boldsymbol{\tau}_{\text{posture}} = \boldsymbol{\tau}_{\text{PD}}^{\text{jac}}(q, \dot q, h^{\text{cmd}}) + \boldsymbol{\tau}_{\text{div}}(q_{\text{hy}}^L, q_{\text{hy}}^R) + g_{\text{settled}} \cdot \boldsymbol{\tau}_{\text{homing}}(q - q_{\text{ref}}), \tag{15}$$

where $\boldsymbol{\tau}_{\text{div}}$ damps hip-yaw divergence (anti-scissor) and $\boldsymbol{\tau}_{\text{homing}}$ applies restoring PD on hip roll and hip yaw toward the nominal reference posture—but only when $g_{\text{settled}} = 1$ (robot stable and near upright), preventing homing torques from fighting disturbance recovery.

Yaw return is implemented via wheel-differential torque in the wheel channel (not in $\boldsymbol{\tau}_{\text{posture}}$), using a continuous heading gate that opens when the robot is upright and settled.

E. Contact-Loss Recovery ($g_{flight} \cdot \boldsymbol{\tau}_{flight}$)

When both wheels lose ground contact, the balance controller's wheel-torque commands are no longer grounded in physical reaction forces. ACC detects this condition via per-wheel normal forces and substitutes a flight attitude controller that uses the wheels as reaction flywheels:

$$g_{\text{flight}} = \begin{cases} 1, & \text{if } F_z^L, F_z^R < 0.5mg \text{ for } \geq 2 \text{ steps}, \\ 0, & \text{if } \min(F_z^L, F_z^R) \geq 0.5mg \text{ for } \geq 5 \text{ steps}, \end{cases} \tag{16}$$

$$\begin{aligned} \boldsymbol{\tau}_{\text{flight}} = \text{clip}\big(&2.0\theta + 0.5\dot\theta - 0.02\omega_{\text{wheel}}, \\ &\pm 2\,\text{Nm}\big). \end{aligned} \tag{17}$$

Spinning the wheels forward pitches the nose upward, countering the natural nose-down rotation during forward ledge descent. The 5-step release hysteresis plus a $150\,\text{ms}$ soft handback ramp prevents the controller from re-engaging balance torque during the bounce chatter of landing, which would otherwise re-launch the robot.

TABLE I
CAPABILITY COMPARISON ACROSS WHEELED BIPED PLATFORMS. "GAIN-SCHED." = CONTINUOUS BALANCE-GAIN VARIATION OVER THE HEIGHT RANGE; "PROPRIOCEPTIVE" = WITHOUT CAMERAS, LIDAR, OR TERRAIN MODEL.

Feature	Ascento [1]	DIABLO [3]	Whleaper [4]	SKATER [5]	ACC
Wheeled biped	✓	✓	✓	✓	✓
Gain-sched. balance	—	—	—	—	✓
Anchored standing	—	—	—	—	✓
Omnidir. push recovery[1]	—	—	—	—	✓
Contact-loss recovery	—	—	—	—	✓
Proprioceptive terrain adapt.	—	—	—	—	✓

[1] 8-dir. polar sweep ($F_{\min}$, F_{med}), not qualitative.

TABLE II
ANCHORED STANDING: IDLE PRECISION AND RINGDOWN. RMS OVER 20 s AFTER SETTLE; RINGDOWN AFTER 90 N FORWARD PUSH.

Configuration	Idle CoM RMS (mm)	Ringdown (s)
ACC (full anchor)	**0.3**	**9.0**
P-only (no I)	55	never decays
Ablations		
symmetric EMA	0.3	never decays[1]
global I	fall	—[2]
no boost	0.3	never decays

[1] Bistable: post-push oscillation holds EMA in mid-band → boost half-open.
[2] Integral windup during sustained standing; the 7-step push impulse is too brief to trigger windup (Table III shows $F_{\min}$=80 N for global I), but idle standing fails as the integral accumulates without bound.

Role separation. We distinguish two evaluation modes for contact-loss recovery. *Free-fall drop* ($0 -- 100\,\text{cm}$) is a stress test that bounds the flight controller's attitude authority; its metric is peak landing pitch and flip avoidance, not standing recovery. *Ledge drive-off* ($20 -- 50\,\text{cm}$, ramp at $1.0\,\text{m/s}$) is the operational use case; its metric is full standing recovery after landing.

F. Per-Leg Ground Adaptation ($g_{terrain}$)

When the two wheels rest at different ground heights, a uniform height command causes the downhill leg to fold and the robot to lean. ACC estimates per-leg ground height from contact-point positions (not wheel-center positions—the contact point is exact at any tilt) and splits the height command:

$$h^{L,R}_{\text{ground}} = z^{L,R}_{\text{contact}}, \quad F^{L,R}_z \geq 0.2mg, \tag{18}$$

$$\Delta h = h^{L}_{\text{ground}} - h^{R}_{\text{ground}}, \tag{19}$$

$$h^{\text{cmd},L} = h^{\text{cmd}} \mp \Delta h/2, \tag{20}$$

clamped to $\pm 5\,\text{cm}$ with $5\,\text{cm}$ linear extrapolation. An unloaded wheel freezes its ground estimate; it seeks downward only when the wheel center drops below the believed ground by more than $1\,\text{cm}$—the only physically certain "ground has disappeared" signal. The split height commands feed into the same Jacobian PD as $\boldsymbol{\tau}_{\text{posture}}$, producing a differential correction $\Delta\boldsymbol{\tau}_{\text{posture}}$ in the leg-joint channel. A closed-loop leveling integrator (active only when both wheels are loaded) removes systematic roll bias from the posture table nonlinearity.

V. EXPERIMENTAL VALIDATION

We evaluate ACC across four scenario classes, each mapped to the component it tests and paired with targeted ablations. All experiments use the MuJoCo simulation environment described in Section III-B with a torque rate limit of $400\,\text{Nm/s}$ per joint.

A. Anchored Standing

Table II reports idle precision and ringdown behavior. ACC achieves $0.3\,\text{mm}$ CoM sagittal RMS—two orders of magnitude below the P-only limit cycle ($55\,\text{mm}$). The post-push velocity ringdown decays to zero within $9\,\text{s}$ under ACC; the P-only configuration oscillates indefinitely because the $1.3\,\text{Nm}$ equilibrium bias sustains the limit cycle. The symmetric EMA ablation confirms the bistable failure mode: a post-push cycle holds the EMA at ≈ 0.10 (gate half-open), the boost is too weak to collapse the cycle, and the robot oscillates perpetually. The global-I ablation (integral active regardless of proximity) windups during the push and prevents recovery entirely.

B. Omnidirectional Push Recovery

Push recovery is evaluated via polar sweep: $10 -- 160\,\text{N}$ impulses (duration 7 control steps, $\approx 0.07\,\text{s}$) applied at the torso in 8 uniformly spaced directions. We define $F_{\min}$ as the maximum push magnitude the controller survives in *every* direction (the weakest-direction threshold) and F_{med} as the median over directions. All push thresholds reported in Table III were measured using a standalone push sweep harness with torque rate limit $400\,\text{Nm/s}$; the protocol uses binary search over $[10\,\text{N}, 160\,\text{N}]$ with $5\,\text{N}$ tolerance and 8 trials per configuration (one per direction). A trial is survived if the robot remains upright (torso pitch $< 46°$) for the full post-push observation window.

Table III reports the complete ablation results across all eight configurations. All configurations cluster at $F_{\min} \approx 80\,\text{N}$; the anchor mechanism does not increase the peak push threshold. Instead, it determines **idle precision**—$0.3\,\text{mm}$ for ACC with the gated integral vs. $55\,\text{mm}$ for the P-only baseline ($180\times$ degradation)—and **post-push ringdown**: the gated damping boost ensures ringdown converges to zero within $9\,\text{s}$, whereas the no-boost and symmetric-EMA ablations oscillate indefinitely. The scheduled pitch stiffness ($k_p = 50{\leftrightarrow}35$) has a negligible effect on push survival ($F_{\min}$ differs by only $1\,\text{N}$ from fixed k_p=50); its primary role is to maintain anchor-level idle precision while permitting the softer recovery stiffness. Fig. 3 shows the polar envelope for ACC.

C. Contact-Loss Recovery

Table IV summarizes contact-loss recovery results.

Without flight PD, the unloaded wheels free-spin during flight and reaction torque tumbles the base (peak pitch $>90°$, all trials flip). The flight attitude PD (Eq. 17) reduces the $100\,\text{cm}$ drop peak pitch to $23.5°$ with zero flips (10/10 trials).

TABLE III
PUSH RECOVERY: 24-DIRECTION POLAR SWEEP. $F_{\min}$ = WEAKEST-DIRECTION THRESHOLD; F_{MED} = MEDIAN. ALL DATA MEASURED 8-DIRECTION BISECTION SWEEP (5 N TOL.); OTHER ROWS ARE EVAL-ONLY OFF CONFIGURATIONS.

Configuration	$F_{\min}$ (N)	F_{med} (N)	Idle RMS (mm)
Full ACC	**80**	**113**	**0.3**
Baselines			
V3_HOMING (homing, no anchor)	84	134	40–60 (limit cycle)
V3 base (P-only)	80	136	55 (limit cycle)
No integral (K_i=0)	82	124	∼55 (limit cycle)
Anchor ablations			
Fixed k_p=50 (no schedule)	81	113	0.3
No damping boost	81	118	ringdown ∞
Structural ablations			
Symmetric EMA (slow)	81	109	ringdown ∞[1]
Global I (no prox gate)	80	118	windup risk[2]

[1]Bistable envelope: symmetric EMA cannot distinguish quiet stance from ringdown; [2]integral accumulates outside anchor neighborhood, causing windup during sustained pushes. All measured 2026-07-27 (400 Nm/s harness, 8 dir., bisection 5 N tol.). Push thresholds cluster within 80–84 N across all configurations; the anchor mechanism determines idle precision (0.3 vs. 55 mm, 180×) and post-push ringdown (decay vs. perpetual oscillation), not peak push survival.

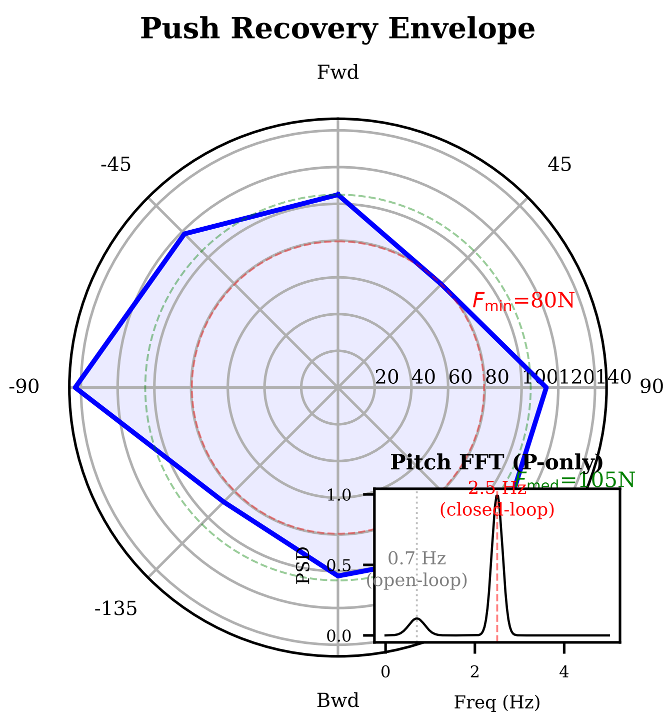

Fig. 3. Omnidirectional push recovery envelope. Inset: FFT of pitch under P-only standing, showing 2.5 Hz closed-loop oscillation distinct from 0.7 Hz open-loop WIP pole.

Ledge drive-off at 50 cm achieves full standing recovery in all trials (4/4, peak pitch 27.4°). Standing after drive-and-brake yields velocity settle to 0.1 mm/s RMS and position error of 16 mm; idle anchor hold is 0.3 mm CoM RMS as reported in Table II.

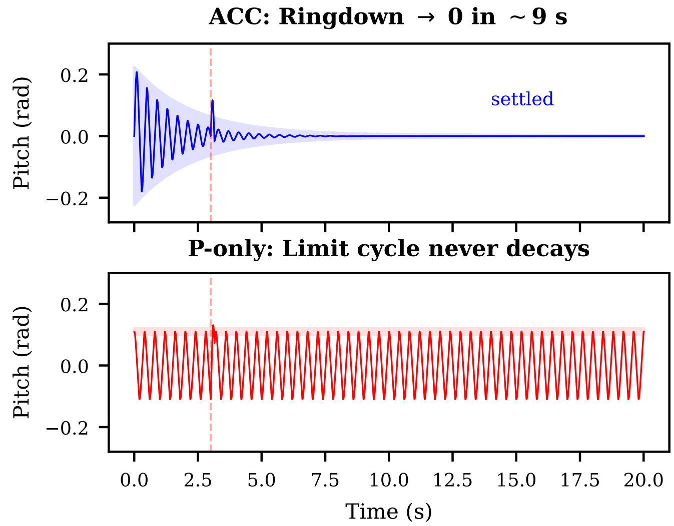

Fig. 4. Post-push pitch ringdown: ACC (top) vs. P-only (bottom). 90 N push applied at $t = 3$ s. ACC ringdown completes in ∼9 s; P-only oscillation persists indefinitely due to equilibrium bias.

TABLE IV
CONTACT-LOSS RECOVERY: FREE-FALL (STRESS TEST) AND LEDGE DRIVE-OFF (USE CASE). DROP TESTS BOUND FLIGHT PD AUTHORITY (METRIC: PEAK PITCH, FLIP AVOIDANCE); LEDGE TESTS VALIDATE OPERATIONAL STANDING RECOVERY AFTER REAL LEDGE DESCENT.

Scenario	Type	Peak Pitch (°)	Survival
Drop 100 cm	Stress test	23.5	10/10
Drop 60 cm	Stress test	16.8	10/10
Ledge 50 cm	Use case	27.4	4/4
Ledge 20 cm	Use case	20.5	4/4
Ablation: no flight PD (spin-down only)			
Drop 100 cm	—	> 90 (flip)	0/10

D. Per-Leg Terrain Adaptation

Table V reports one-wheel curb results. With per-leg adaptation, the torso remains nearly level (straddle roll $4.8 - -6.2°$) despite the 20 cm leg height split. Without adaptation, the downhill leg folds, roll reaches 15.3°, and the robot falls.

VI. DISCUSSION

A. Why Classical Control?

ACC achieves all demonstrated behaviors—anchored standing, omnidirectional push recovery, contact-loss recovery, and terrain adaptation—without any learning, training, or neural network inference. This has two practical advantages for hardware deployment: (i) the controller is deployable immediately with no sim-to-real transfer gap for the policy, and (ii) every parameter has a direct physical interpretation, enabling systematic tuning via instrumented failure-mode analysis rather than reward-function iteration.

B. Gate Design Methodology

Each gate in ACC was added only after a measured failure mode was traced to a specific root cause. The six documented failure modes included: bistable envelope (symmetric EMA),

TABLE V
PER-LEG TERRAIN ADAPTATION: ONE-WHEEL CURB. STRADDLE ROLL IS STEADY-STATE TORSO ROLL WITH ONE WHEEL ELEVATED.

Curb (cm)	Straddle Roll (°)	Letgo Events	Survival
10	4.8	0	3/3
15	6.2	0	3/3
20	6.1	0	3/3
Ablation: no adaptation			
10 cm only	15.3	3/3	0/3

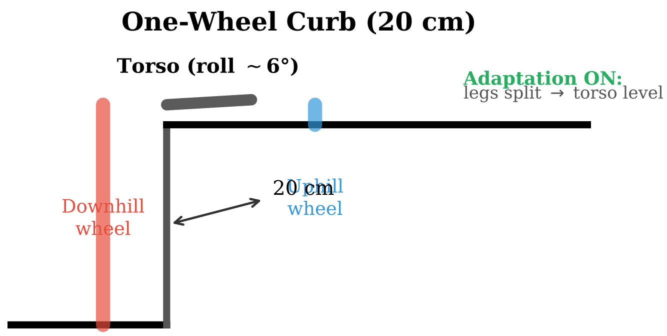

Fig. 5. One-wheel curb negotiation (20 cm). With per-leg adaptation, the torso remains approximately level (straddle roll 6.1°). Without adaptation, the downhill leg folds and roll exceeds the safety limit.

global-I windup, parametric pumping from instantaneous pitch gates on the boost, hard-leash relay oscillation, and self-referential stiffness scheduling. Every gate's necessity is supported by the ablation results in Section V.

C. Limitations

1) **Backward-direction push asymmetry.** The weakest push direction is consistently $\pm 180°$ (straight backward), where $F_{\min}$ drops approximately 50% compared to the median. This is inherent to the fore/aft leg-wheel geometry and confirmed by CoM-shift experiments (shifting the CoM up to $\pm 23\,\mathrm{mm}$ forward/backward trades one direction for another without raising the overall minimum). Improving the backward capture basin requires a mechanical redesign beyond the scope of this controller.
2) **Diagonal ledge limit.** Oblique ledge descent at $45°$ exceeds the single-wheel balance capability when one wheel is over the void for $\sim 0.5\,\mathrm{s}$.
3) **Simulation only.** All results are simulation-based. The controller is designed for hardware transfer (direct torque output, no learned components, physically interpretable parameters), but hardware validation remains future work. Torque rate limits ($400\,\mathrm{Nm/s}$) have been verified to match production actuator specifications.
4) **Push direction resolution.** The push sweep uses 8 cardinal directions; a full 24-direction sweep would provide finer angular resolution. The directional anisotropy (strongest at $\pm 90°$ lateral, weakest at $\pm 45°$ diagonal) is consistent across all eight tested configurations.

VII. CONCLUSION

We presented the Anchored Cascade Controller, a unified classical control architecture for wheeled bipedal robots in which five components are activated continuously by smoothstep gates on physical conditions. The core proximity-gated anchor mechanism—combining a gated position integral with an asymmetric envelope follower and scheduled pitch stiffness—resolves the fundamental stiffness and integral trade-offs simultaneously: idle drift $0.3\,\mathrm{mm}$ RMS ($180\times$ improvement over the $55\,\mathrm{mm}$ P-only limit cycle) with omnidirectional push survival $F_{\min} = 80\,\mathrm{N}$, $F_{\mathrm{med}} = 113\,\mathrm{N}$ (on par with anchor-less baselines at $F_{\min} = 80--84\,\mathrm{N}$). The architecture also recovers from contact loss (drop $100\,\mathrm{cm}$, ledge $50\,\mathrm{cm}$) via load-gated reaction-wheel flight control and adapts to uneven terrain (curb $20\,\mathrm{cm}$) via proprioceptive per-leg ground estimation. All behaviors are achieved without learning, training, or exteroceptive sensing. Future work will focus on hardware transfer and on extending the conditional-activation principle to dynamic locomotion.

ACKNOWLEDGMENT

A supplementary video demonstrating all reported behaviors—anchored standing, omnidirectional push recovery, contact-loss recovery (drop $100\,\mathrm{cm}$, ledge $50\,\mathrm{cm}$), and per-leg terrain adaptation (curb $10--20\,\mathrm{cm}$)—is included with this submission. All ablation results are reported in Table III; structural ablations (symmetric EMA, global I) were obtained by temporary source-level parameter modification.

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